

Roman Stepaniuk

Frontend Developer

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English (B1 – Intermediate) • Ukrainian (Native)

[Telegram](#) • [GitHub](#) • [Portfolio](#) • [LinkedIn](#)

Frontend Engineer with 2.5+ years of experience in building complex, high-performance web and desktop applications using the React ecosystem. Specialized in scalable architecture, strict type safety, and solving engineering challenges that bridge the gap between UI, hardware (Electron), and backend systems.

WORK EXPERIENCE

Frontend Developer | PAM (Event Tech / Parking Automation) | Remote

Feb 2024 – present

Project: "Tviy Kvytok" (Event Ticketing Platform)

Led frontend development for a high-traffic events platform, establishing engineering culture and optimizing core flows for **2,500+ daily active users**.

- Decoupled monolithic booking module via React Adapter Pattern + Zustand – eliminated layout thrashing and unnecessary re-renders
- Built canvas-based seat-selection engine with R-tree spatial indexing and viewport virtualization – smooth zoom/pan across complex venues, real-time seat-lock via SSE
- Engineered browser barcode scanner (ZXing) with custom heuristic for ultra-wide lens detection – +40% scan success rate on low-end devices
- Built high-performance calendar (react-day-picker + react-virtuoso windowing) – significantly reduced TBT on mobile
- Architected type-safe multi-step checkout (RHF + Zod) with Optimistic UI – instant feedback under race conditions
- Architected a robust quality ecosystem (CI/CD, Static Analysis, Datadog). Enforced strict Code Reviews and mentored the team, cutting regression bugs by ~40%.

Technical Stack: Next.js, TypeScript, TanStack Query, React Hook Form, Zod, Recharts, React Virtuoso, ZXing, Seats.io, Orval, Shadcn/UI

Project: Project: Parking Automation Ecosystem (Electron kiosk + Admin panel)

Architected a complete frontend ecosystem for a parking platform processing **1M+ UAH monthly**, 75K+ sessions, zero downtime. Built desktop apps (Electron kiosk, cashier terminal), PWA admin panel, and deep hardware integration layer.

- Engineered scalable Redux architecture with dynamic slice injection – **reduced bundle by 40%**, enabling instant startup
- Built custom Redux Listener Middleware orchestrating WebSockets + Electron IPC – eliminated resource leaks in multi-process architecture
- Architected fault-tolerant I/O layer (COM/serial) in isolated Node.js child processes – 100% UI responsiveness
- Reverse-engineered CCNET and CashCode protocols – implemented binary packet layer (SYNC framing, CRC-16), polling state machine with Escrow/Stack flow and real-time payment session tracking
- Built B-POS terminal driver – binary protocol (STX/LRC), async command queue with ACK/NAK/EOT flow control and scheduled Z-Balance via cron

Technical Stack: React, Electron, Redux Toolkit, TypeScript, TanStack (Query, Router), React Hook Form, Recharts, Orval, Shadcn/UI, Winax, Serialport

Built a white-label Next.js checkout solution with Shopify integration, enabling merchants to customize branding, payments, and analytics without code deployments.

- Built runtime config layer for white-label checkout – UI, payments (PayPal/Klarna), GTM customizable without deploys
- Engineered SSR flows with intelligent caching strategies, achieving sub-second load times and optimal Core Web Vitals scores
- Implemented i18n (react-intl) with auto locale detection and locale-aware routing
- Developed a complex payment form with multi-step validation using React Hook Form and Zod

Technical Stack: Next.js, TypeScript, TanStack Query, React Window, React Intl, React Hook Form, Zod, Shadcn/UI

EDUCATION

National Aviation University, Kyiv

2023 - present

Faculty of Computer Science and Information Technology

TECHNICAL SKILLS

Languages: TypeScript, JavaScript (ES6+), HTML5, CSS3 (Sass/SCSS)

Frontend: React, Next.js, Redux Toolkit (RTK Query), TanStack (Query, Router), React Hook Form, Zod, Zustand, Orval, React Window / Virtuoso, Tailwind CSS, Shadcn/UI, Framer Motion, Recharts, React-Konva, MUI, Ant Design

Backend: NestJS, Electron, WebSockets, Prisma ORM, Type ORM, PostgreSQL, Firebase, SerialPort

Tooling: Vite, Docker, Git, CI/CD, Jest